REDWINE
QUALITY PREDICTOR

RED WINE QUALITY PREDICTION

Introduction:

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions without being explicitly programmed. In this project, we predict the quality of red wine based on its chemical properties using machine learning algorithms.

The Red Wine Quality dataset contains several chemical features such as acidity, sugar, pH, alcohol content, and sulphates. Based on these features, the quality of wine is predicted.

This project helps in understanding classification problems and comparing the performance of two machine learning algorithms.

Objective

The main objective of this project is:

- To predict red wine quality using machine learning
- To apply two different ML algorithms
- To compare the performance of both algorithms
- To find the best algorithm for accurate prediction

Dataset Description

The dataset used in this project is the Red Wine Quality dataset from Kaggle.

Dataset Features:

- Fixed acidity
- Volatile acidity
- Citric acid
- Residual sugar
- Chlorides
- Free sulfur dioxide
- Total sulfur dioxide
- Density
- pH
- Sulphates
- Alcohol
- Quality (Target Column)

Dataset Type:

CSV File

Total Records:

1599 rows

Total Columns:

12 columns

Algorithms Used

Algorithm 1: Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of a class label.

Advantages:

- Simple and fast
- Easy to understand
- Good for binary classification

Algorithm 2: Random Forest Classifier

Random Forest is an ensemble learning algorithm that uses multiple decision trees and combines their outputs for better prediction.

Advantages:

- High accuracy
- Handles large datasets well
- Reduces overfitting

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook / VS Code

Methodology

Step 1: Import Dataset

The CSV file is loaded using Pandas.

Step 2: Data Preprocessing

- Check missing values
- Understand dataset structure
- Split input and output columns

Step 3: Train-Test Split

The dataset is divided into:

- Training data (80%)
- Testing data (20%)

Step 4: Apply Algorithms

Two algorithms are trained:

- Logistic Regression
- Random Forest Classifier

Step 5: Model Evaluation

Accuracy score is calculated for both models.

Step 6: Comparison

Both algorithm results are compared to identify the better model.

Results

After training both models:

Logistic Regression Accuracy

Around 60% to 65%

Random Forest Accuracy

Around 70% to 80%

Random Forest performs better than Logistic Regression for this dataset.

Conclusion

This project successfully predicts red wine quality using machine learning techniques. Two algorithms were applied and compared.

Random Forest Classifier showed better performance compared to Logistic Regression due to its higher accuracy and better handling of complex data patterns.

This project demonstrates the practical implementation of machine learning in real-world quality prediction systems.